Mathematical Memoirs

A Personal Record of Discoveries, Speculations, and Proofs

Maintained by: **Eric Asaah**
Project: *The SAT Shortcut Series*

"The best way to understand a theorem is to discover it yourself."
— Pólya

Entry 01: The Transversal Invariant

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1. The Starting Point

It started with a single, unremarkable SAT question.

A transversal intersects two parallel lines, forming eight angles. One of those angles is $(7x-250)^\circ$. The question asks: *which of the following could* ***not*** *be the sum of four of these angles?*

A) $-14x+1540$ **B)** $14x-320$ **C)** $-28x+1720$ **D)** 360

All answer choices are linear expressions in x (or constants), so they can be treated as functions of x that must agree with some possible sum for every x if they are to be valid.

The standard approach is to recognise that every angle is either $a = (7x-250)^\circ$ or its supplement $b = (-7x+430)^\circ$, and then carefully test all five possible combinations of four angles. It works. But it is slow, error-prone, and feels more like bookkeeping than thinking.

I became suspicious that something cleaner was hiding underneath.

2. The Thought Process

Parametrising the Procedure. I got the question wrong first. After I looked at the solution, I followed the standard approach: call one angle a, the other $b = 180-a$, then list all five combinations that give four angles: $4a$, $3a+b$, $2a+2b$, $a+3b$, $4b$. Substitute in and compare to the answer choices. It works. But I am very lazy, and doing this four or five times per question was not happening.

So I started parametrising. I substituted $b = 180-a$ directly into each combination and simplified:

$$4a, \quad 2a+180, \quad 360, \quad 540-2a, \quad 720-4a.$$

Then I noticed all five fit a single expression. If I let $m \in \{0,1,2,3,4\}$ count how many type-a angles I pick, every valid sum is:

$$K = (2m-4)\,a + (4-m)\cdot 180.$$

The 4 here was specific to this question. It later became n when the next question of the same type asked for two angles instead of four — that forced me to generalise:

$$K = (2m-n)\,a + (n-m)\cdot 180.$$

Now I could see that the x-coefficient in any valid answer comes entirely from the $(2m-n)\cdot k$ factor. Written out, the coefficient of x is $C = (2m-n)k$, and as m runs from 0 to n the values form an arithmetic progression $-nk, -(n-2)k, \ldots, nk$ with step $2k$. For this question with $k = 7$ and $n = 4$, the valid x-coefficients are -28, -14, 0, 14, 28. I derived a formula to recover m from any candidate coefficient C:

$$m = \frac{1}{2}\left(\frac{C}{k} + n\right).$$

But when I tested this against the four choices, every single one produced a valid integer m. The College Board had constructed the question so that the x-coefficients alone would not eliminate anything. I needed more.

The Desmos Mess and the Accidental Observation. My instinct was to move to Desmos. I plotted the general formula with a slider for m and all four answer choices on the same graph. (x on the horizontal axis, the sum expression on the vertical.) The screen immediately became unreadable — lines everywhere, no way to tell which was which. I cleared everything and tried again.

This time I replaced the slider with a list, writing the entire bundle of valid sums as a single expression:

$$y = (2\cdot[0\ldots4] - n)\,a + (n - [0\ldots4])\cdot 180.$$

and then later, this:

$$y = (2i - n)\,a + (n-i)\cdot 180 \quad for \quad i = [0\ldots n].$$

Desmos drew all valid lines at once. I then plotted the four SAT choices on top. Three of them landed exactly on bundle lines. One did not. That was enough to answer the question — but while I was staring at the graph, I noticed something I had not been looking for: all the bundle lines seemed to be passing through one common point. I almost ignored it. I was still focused on the m formula and whether the constants could give me another constraint. The common point just sat there while I chased other things.

The Visual Anomaly. I zoomed out further and the convergence became undeniable. Every single valid sum equation, regardless of how many type-a and type-b angles it contained, passed through one fixed coordinate. I zoomed in and confirmed it. This was not expected, and I started asking why.

The Proof. I rewrote the sum to separate the role of m:

$$S_m = ma + (n-m)(180-a) = m(2a-180) + n(180-a).$$

Now m enters only through the factor $(2a - 180)$. When that factor is zero, the choice of m no longer matters: all the expressions give the same value. Setting $2a - 180 = 0$ gives $a = 90°$, and then any sum of n angles becomes $n \cdot 90°$ no matter how many a's and b's are picked:

$$S_m = 90(2m-n) + 180(n-m) = 180m - 90n + 180n - 180m = 90n.$$

Therefore every line in the bundle must pass through the point where the angle equals 90° and the sum equals $90n$. Translating $a = 90$ back to x using $a = kx + t$ yields

$$x_p = \frac{90 - t}{k}, \qquad y_p = 90n.$$

I verified this against the SAT choices. Choice A gave 860 at that point. Every other choice gave $360 = 90 \times 4$. The answer was confirmed.

3. The Intermediate Machinery — What Almost Worked

Before the invariant point appeared, there were several earlier attempts to build a clean filter. They are recorded here because they each captured something real, but none was complete on its own.

Attempt 1 — Filtering by the x-coefficient alone. The master sum formula gives the coefficient of x in any valid sum as $(2m - n) \cdot k$. Given a candidate expression $Cx + T$, recover m as:

$$m = \frac{1}{2}\left(\frac{C}{k} + n\right).$$

If this is not an integer, the choice is immediately impossible. This felt powerful. But the SAT constructs its trap choices so that m is always an integer — every wrong answer has a plausible x-coefficient. The filter killed nothing in practice.

Attempt 2 — Adding a parity check. The term $\frac{C}{k} + n$ must be even for m to be an integer. If n is even, then C/k must be even; if n is odd, C/k must be odd. This is sometimes faster to check than computing m explicitly. But again, the College Board ensures all choices pass it.

Attempt 3 — Using m to pin the constant T. Once m is found from the x-coefficient, it pins the constant too. Every valid sum satisfies:

$$T = m \cdot t + (n - m)(180 - t).$$

The full check is: compute m from C, then verify T against this formula. This is rigorous and correct. The problem is that it still requires computing m for each of the four choices and doing sign-heavy arithmetic on the constant every time. Reliable, but not fast.

Attempt 4 — The mod 180 idea (discarded). I considered whether T could be filtered using $T \bmod 180$. The intuition: each swap of an a for a b changes the constant by exactly $180 - 2t$, so perhaps T lives on a predictable residue class. This was a dead end. The step size $180 - 2t$ is not generally a divisor of 180, so the residues are irregular and produce no clean criterion.

What all of these had in common. Every approach above was a test applied *per choice*, requiring separate arithmetic for each of the four options. The invariant point is computed *once* from the problem setup and then all four choices are tested by simple substitution. The shift from "check each m" to "check one x" is what made the final method genuinely fast.

4. The Fundamental Theorem

Theorem (Transversal Invariant Point).
Let a transversal intersect any number of parallel lines, producing angles of measure $a(x) = kx + t$ with $k \neq 0$. For a fixed integer n (number of angles summed), define $S_m(x) = m \cdot a(x) + (n - m)(180° - a(x))$ for $m = 0, \ldots, n$. Then every such expression satisfies

$$S_m\left(\frac{90 - t}{k}\right) = 90n.$$

5. Proof of Universality

The pool of angles in any parallel-transversal system is always $\{a,\ 180 - a\}$, regardless of how many parallel lines are present. Adding more parallel lines adds more angle pairs, but never introduces a third type. Thus:

$$S_m = m \cdot a + (n - m)(180 - a) = m(2a - 180) + n(180 - a).$$

The coefficient of m is $2a - 180$. For the expression to be independent of m, we need $2a - 180 = 0$, i.e. $a = 90°$. Setting $kx + t = 90$ gives $x_p = \dfrac{90 - t}{k}$, and evaluating any S_m there yields

$$S_m(x_p) = 90(2m - n) + 180(n - m) = 90n.$$

Thus every possible sum line passes through $\left(\frac{90-t}{k},\ 90n\right)$. $\square$

The theorem holds for all n (number of angles summed) and all L (number of parallel lines). Both are irrelevant to the structure of the proof.

6. Preliminary Filters

A valid sum expression $Cx + T$ must satisfy three conditions. The Invariant Test is the last and most decisive:

1. **Integer Constraint:** $m = \dfrac{1}{2}\left(\dfrac{C}{k} + n\right)$ must be an integer.
2. **Range Constraint:** $0 \leq m \leq n$.
3. **Invariant Test:** $C\left(\dfrac{90 - t}{k}\right) + T = 90n$.

In practice on the SAT, the College Board ensures all choices pass Conditions 1 and 2, deliberately forcing you to reach the algebra. Condition 3 is what cuts through it. (Passing the invariant test is necessary but not sufficient in general; on this test the distractors are built so that exactly one choice fails it, making the test decisive.)

7. Strategic Application

To solve a "Which could NOT be k" question in under ten seconds:

1. Solve for x where the given angle $= 90°$: $\quad x_p = \dfrac{90-t}{k}$.
2. Compute the target: $\quad y_p = 90n$.
3. Substitute x_p into each answer choice.
4. The choice that does not equal y_p is the answer.

8. Verification Drills: SAT Format

The following five questions were used to pressure-test the theorem. All were solved correctly using only the Invariant Point.

The answer choices are linear expressions. If an expression were truly equal to some valid sum for all x, the two functions would have to agree at every x. Checking them at a single x is therefore enough to expose an impostor.

Question 1. A transversal line intersects two parallel lines, forming eight angles. The measure of one of these angles is $(7x-250)°$. The sum of the measures of four of the eight angles is $k°$. Which of the following could **NOT** be equivalent to k, for all values of x?

A) $-14x+1540$ **B)** $14x-320$ **C)** $-28x+1720$ **D)** 360

Pivot: $7x-250=90 \Rightarrow x=\frac{340}{7}$. Target: 360.
Choice A: $-14\cdot\frac{340}{7}+1540=-680+1540=860\neq 360$. **Answer: A.**

Question 2. In the xy-plane, a transversal intersects two parallel lines. The measure of one of the eight angles formed is $(12x+30)°$. If k is the sum of the measures of two of these angles, which of the following could **NOT** be equivalent to k?

A) $24x+60$ **B)** 180 **C)** $-24x+330$ **D)** $12x+150$

Pivot: $x=5$. Target: 180.
Choice D: $12(5)+150=210\neq 180$. **Answer: D.**

Question 3. A line intersects two parallel lines. The measure of one of the angles formed is $(2x-50)°$. If the sum of the measures of six of the angles is $k°$, which of the following could **NOT** be equivalent to k?

A) $12x-300$ **B)** 540 **C)** $4x+340$ **D)** $8x+40$

Pivot: $x=70$. Target: 540.
Choice C: $4(70)+340=620\neq 540$. **Answer: C.**

Question 4. The measure of one of the eight angles formed by a transversal and two parallel lines is $(10x-110)°$. The sum of the measures of four of these angles is $k°$. Which of the following could **NOT** be equivalent to k?

A) $20x + 140$ **B)** 360 **C)** $40x - 440$ **D)** $20x + 200$

Pivot: $x = 20$. Target: 360.
Choice A: $20(20) + 140 = 540 \neq 360$. **Answer: A.**

Question 5. Two parallel lines are intersected by a transversal. The measure of one of the angles is $(5x + 15)^\circ$. If k is the sum of the measures of three of these eight angles, which of the following could **NOT** be equivalent to k?

A) $15x + 45$ **B)** $5x + 345$ **C)** $5x + 205$ **D)** $-5x + 375$

Pivot: $x = 15$. Target: 270.
Choice B: $5(15) + 345 = 420 \neq 270$. **Answer: B.**

Key Insight: The Invariant Point Filter

To check whether a candidate expression $Cx + T$ can represent a sum of n angles from a parallel-transversal system where one angle is $(kx + t)^\circ$,

1. Compute $x_p = \dfrac{90 - t}{k}$ (the x that makes the angle 90°).
2. Compute the required value $y_p = 90n$.
3. Evaluate the candidate at x_p: $Cx_p + T$.

If the result is not y_p, the candidate is impossible.

End of Entry 01.

Theorem holds for all n and any number of parallel lines.